

Shadab Zafar

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Work

Jun'2019 – **Software Engineer III**, *Tower Research Capital*, Gurgaon

- Present
- Designed & implemented a C++ based real-time risk analytics engine processing over 200K updates/sec on the entire firm's portfolio. Leveraged an in-memory DuckDB store to provide a live SQL API & ClickHouse for historical timeseries based usecases.
 - Improved performance of a C++ based trade monitoring application by multi-threading & removing lock-contention resulting in reduced web API response times from ~100ms to ~10ms.
 - Owned & maintained an anomaly detection framework that aggregates live trade metrics & alerts when they deviate from historical averages stored in a multi terabyte RocksDB instance.
 - Created a C++ based autocomplete & search API that powers a trading frontend application.
 - Managed various web-based Risk Monitoring applications written in Python/Django/Flask etc.
 - Consolidated ad-hoc scripts & data analysis tools into a maintainable Python/Streamlit dashboard.

Open Source Internships

- Jun – Aug 2016 **Student Developer**, *Google Summer of Code*, The HoneyNet Project
Python 2 to 3 port and improved test coverage of an HTTP proxy tool (mitmproxy)
- Jun – Aug 2015 **Student Developer**, *Google Summer of Code*, MetaBrainz Foundation
Improvements to a Python/Flask based music review website (CritiqueBrainz)
- Jun – Aug 2014 **Student Developer**, *Google Summer of Code*, MetaBrainz Foundation
Created a website & a Python based plugin API for a music tagging tool (Picard)

Projects

- 2018 – 2019 **Bitcoin**, C++, Python
Masters thesis work involved modifying Bitcoin Core code to implement approaches of improving chain stability. Created tools to manage a testbed of Bitcoin nodes allowing us to run simulations on an internal cloud cluster. Paper published in COMSNETS 2022.
- 2017 – 2018 **Blockchain**, Python, Javascript, Solidity
Masters coursework included an Ethereum smart contract and a simulator for blockchain growth.
- Cheat**, Go
A tool that allows you to create and view interactive cheatsheets on the command-line.
- Automation Scripts**, Python, Bash, AutoHotkey
Like writing scripts to automate tasks like: renaming files, capturing & uploading screenshots etc.

Education

- 2017 – 2019 **M.Tech**, *Computer Engineering*, Indian Institute of Technology Delhi, GPA: 9.2/10
- 2012 – 2016 **B.Tech**, *Computer Engineering*, Jamia Millia Islamia, New Delhi, GPA: 9.24/10

Publications

- 2022 **COMSNETS**, Improved Stability & Performance of PoW Blockchains using Anchors

Tech Talks

- Mar 2016 **FOSSASIA**, Singapore, Conference talk on command-line applications in Python